

PINN for ODE Cox Model

1. Cox Model for constant covariates x has the following form:

$$\lambda_x(t) = \alpha(t) \exp(x^T \beta) \quad (1)$$

We can write this as an ODE:

$$\Lambda'_x(t) = \alpha(t) \exp(x^T \beta) \quad (2)$$

In order to enforce positivity of baseline hazard, define $\gamma(t) = \log \alpha(t)$.

$$\Lambda'_x(t) = \exp(\gamma(t)) \exp(x^T \beta) = \exp(x^T \beta + \gamma(t))$$

And initial condition: $\Lambda_x(0) = 0$

(3)

2. Neural Architecture

Since the $\gamma(t)$ coefficient is a time dependent function, we need a separate network to approximate this function. Then, we have:

- **Surrogate network** (Λ_ϕ): takes time t and covariates x as inputs, and outputs $\hat{\Lambda}_\phi(t, x)$.
- **Coefficient network** (γ_θ): takes time t as input and outputs $\hat{\gamma}_\theta(t)$.
- β **vector** of learnable parameters.

3. Loss function: $L_{\text{total}} = w_1 L_{\text{MLE}} + w_2 L_{\text{PL}} + w_3 L_{\text{ODE}} + w_4 L_{\text{IC}}$

- L_{MLE} : negative log-likelihood (grounding γ and Λ networks):

$$L_{\text{MLE}} = -\frac{1}{n} \sum_{i=1}^n \left[\Delta_i \{ \hat{\gamma}_\theta(Y_i) + x_i^T \beta \} - \hat{\Lambda}_\phi(Y_i, x_i) \right] \quad (4)$$

- L_{PL} : negative log partial likelihood (auxiliary loss for β stabilization):

$$L_{\text{PL}} = -\frac{1}{n_e} \sum_{i: \Delta_i=1} \left[x_i^T \beta - \log \left(\sum_{j \in R(Y_i)} \exp(x_j^T \beta) \right) \right] \quad (5)$$

where n_e is the number of events and $R(Y_i) = \{j : Y_j \geq Y_i\}$ is the risk set at time Y_i .

- $L_{\text{ODE}} = \frac{1}{N} \sum_{j=1}^N \left(\frac{\partial \hat{\Lambda}_\phi(t_j, x_j)}{\partial t} - \exp(\hat{\gamma}_\theta(t_j) + x_j^T \beta) \right)^2$

where N are **collocation points**.

- L_{IC} : initial condition loss, enforcing condition at $t = 0$:

$$L_{\text{IC}} = \frac{1}{n} \sum_{i=1}^n \left(\hat{\Lambda}_\phi(0, x_i) \right)^2 \quad (6)$$